Universidad de las Fuerzas Armadas ESPE

Software Engineering — Advanced Web Development

Features Document

Project: Biconoir's Restaurant

Team: ByteClub

Feature 1: Customer UI & Digital Menu Experience

This feature covers the full customer-facing interface, brand identity, menu browsing, ingredient customization, shopping cart management, and the order checkout flow.

- **Task 1.1:**
  Set up the base web application template with a global Spanish language localization (es-ES) applied across all views, navigation menus, alerts, and button labels throughout the system.
- **Task 1.2:**
  Implement the Biconoir brand identity: integrate the green logo in the header, apply the #1a4731 green color palette via CSS and Tailwind across all pages, and ensure proper contrast and readability for all UI elements.
- **Task 1.3:**
  Develop the dynamic menu page using a card-based grid layout. Each card displays the dish name, product image, price badge, and a short description. The grid responds to 1/2/4 column breakpoints.
- **Task 1.4:**
  Build the ingredient customization modal: when a user clicks 'Add to Cart', a modal opens showing the dish image, price, description, checkboxes for each ingredient, and a quantity selector (1–10). The user can deselect ingredients before confirming.
- **Task 1.5:**
  Implement the shopping cart view with the ability to add items (with custom ingredients), remove individual items, display quantity per item, and calculate the running total. On checkout, the cart is cleared and a new Order record is persisted to the database.
- **Task 1.6:**
  Build the user authentication flow: registration form (name, phone, email, password) with frontend validation (min 6-char password, phone format, .com email), and a login form with credential verification against the database. Sessions are maintained across the application.

Feature 2: Table Reservation Management

This feature handles the full reservation lifecycle, from the customer-facing booking UI with a live availability calendar, through backend capacity validation, to the administrator's reservation management dashboard.

- **Task 2.1:**
Create the customer-facing reservation form capturing: full name, date (selected from a 14-day interactive calendar), time slot (30-minute intervals, 12:00–22:30), number of guests (1–20), experience type (Standard / Special Event / Business / Wine Tasting), duration (1–4 hours), and optional notes for allergies or preferences.

- **Task 2.2:**
Implement backend capacity validation: each time slot supports a maximum of 3 concurrent reservations. If a slot is full, the UI marks it as occupied. Duration-aware blocking prevents overlapping bookings within the same time window. A user may hold at most 2 active reservations total and 1 per day.

- **Task 2.3:**
Build the 'My Reservations' tab on the customer reservation page, listing all active reservations (date, time, pax, duration) with a cancel button. Cancellation updates the status to 'Cancelled' and instantly frees the slot.

- **Task 2.4:**
Build the Administrator Reservations dashboard: displays all reservations in a sortable table (ID, date/time, pax, customer name, notes, status). Includes a date filter and a real-time search bar that filters by ID, name, or date without a page reload.

Feature 3: Operational Backend & Inventory Workflow

This feature covers the internal operational mechanics: inventory management with batch tracking, automatic stock deduction on order completion, order status management, customer satisfaction surveys, and the admin control panel.

- **Task 3.1:**
Develop the Inventory module: administrators can view all ingredients with their SKU, current available stock, unit of measurement, and status badge (OPTIMAL / LOW STOCK / OUT OF STOCK). A search bar filters the inventory table in real time.

- **Task 3.2:**
Implement the 'Load Supplies' form to register new inventory batches: select or create an ingredient by name, set unit of measurement, quantity, cost per unit, and purchase date. Stock totals are recalculated automatically via a syncStock() operation after each batch insert.

- **Task 3.3:**
Implement automatic ingredient stock deduction: when an admin marks an order as 'Completed', the system iterates through each order detail, retrieves the dish's required ingredient quantities, and deducts from batches in FIFO order. If stock is insufficient, the operation is rolled back and an error message is shown.

- **Task 3.4:**
Build the Order Management section in the admin dashboard: displays a filterable, date-filtered table of orders with ID, customer, item breakdown, total, and status. Admins can mark orders as 'Completed' (triggers stock deduction) or 'Cancelled' via confirmation dialogs.

- **Task 3.5:**
Implement the Dish Management CRUD: admins can add new dishes (name, description, price, image URL, ingredient composition with required quantities), edit existing dishes, and soft-delete dishes (is_available = false). Business rules prevent editing or deleting dishes with active pending orders.

- **Task 3.6:**
Build the Customer Satisfaction Survey module: customers fill a public survey form (name, 1–5 star rating, comment) with frontend validation. The admin panel displays all submitted surveys in a searchable table sorted by most recent, showing customer name, star rating, comment, and submission date.